EXPERIMENT 19

IMPLEMENT CONTINUOUS DEPLOYMENT USING GITHUB ACTIONS TO DEPLOY A DOCKERIZED APPLICATION

Aim

To implement Continuous Deployment using GitHub Actions to automatically build, push, and deploy a Dockerized application.

Requirements

GitHub account, Docker installed, Docker Hub account, a cloud VM/server with Docker installed, and basic Git knowledge.

Procedure

1. Set up GitHub Repository

Create a GitHub repository for the Dockerized application. Initialize the local project with Git, add the application files, commit the changes, and push them to the main branch.

2. Create and Containerize the Application

Create a simple application and a Dockerfile. Build the Docker image using **docker build** and verify it locally by running a container.

3. Create GitHub Actions Workflow

Create a workflow file under **.github/workflows/**. Configure the workflow to trigger whenever new code is pushed to the main branch. The workflow checks out the code, builds the Docker image, pushes it to Docker Hub, and deploys the latest image to the cloud server.

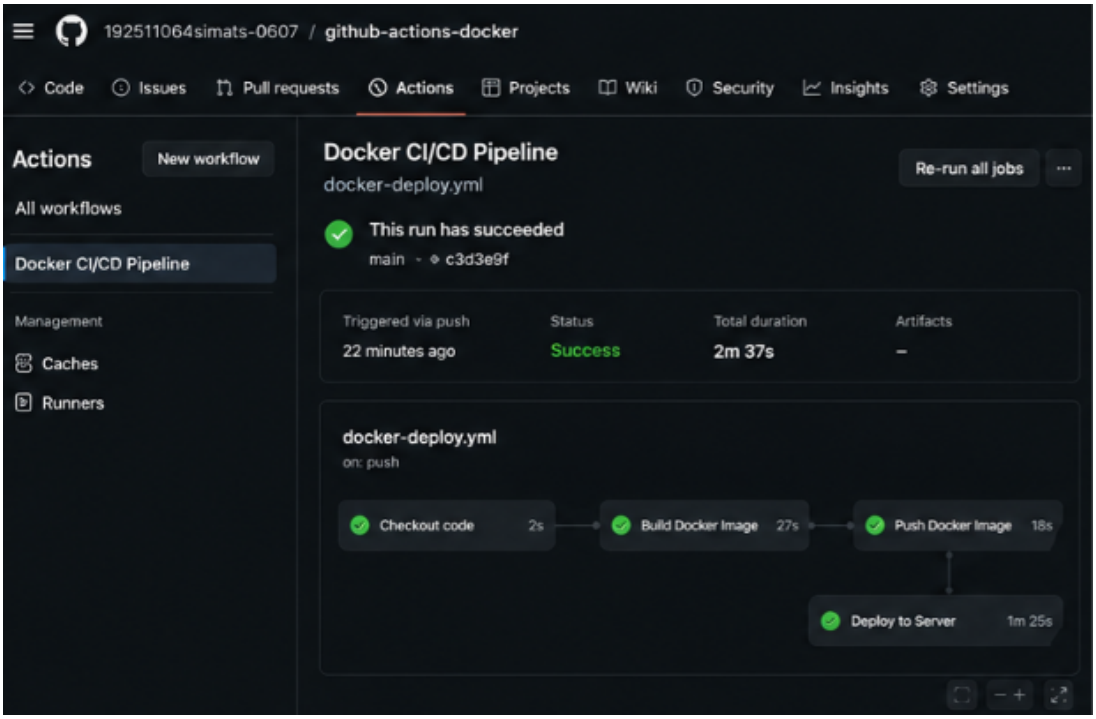
4. Configure Secrets

Add the required Docker Hub and cloud-server credentials as GitHub Actions repository secrets. Sensitive credentials should never be written directly in the workflow file.

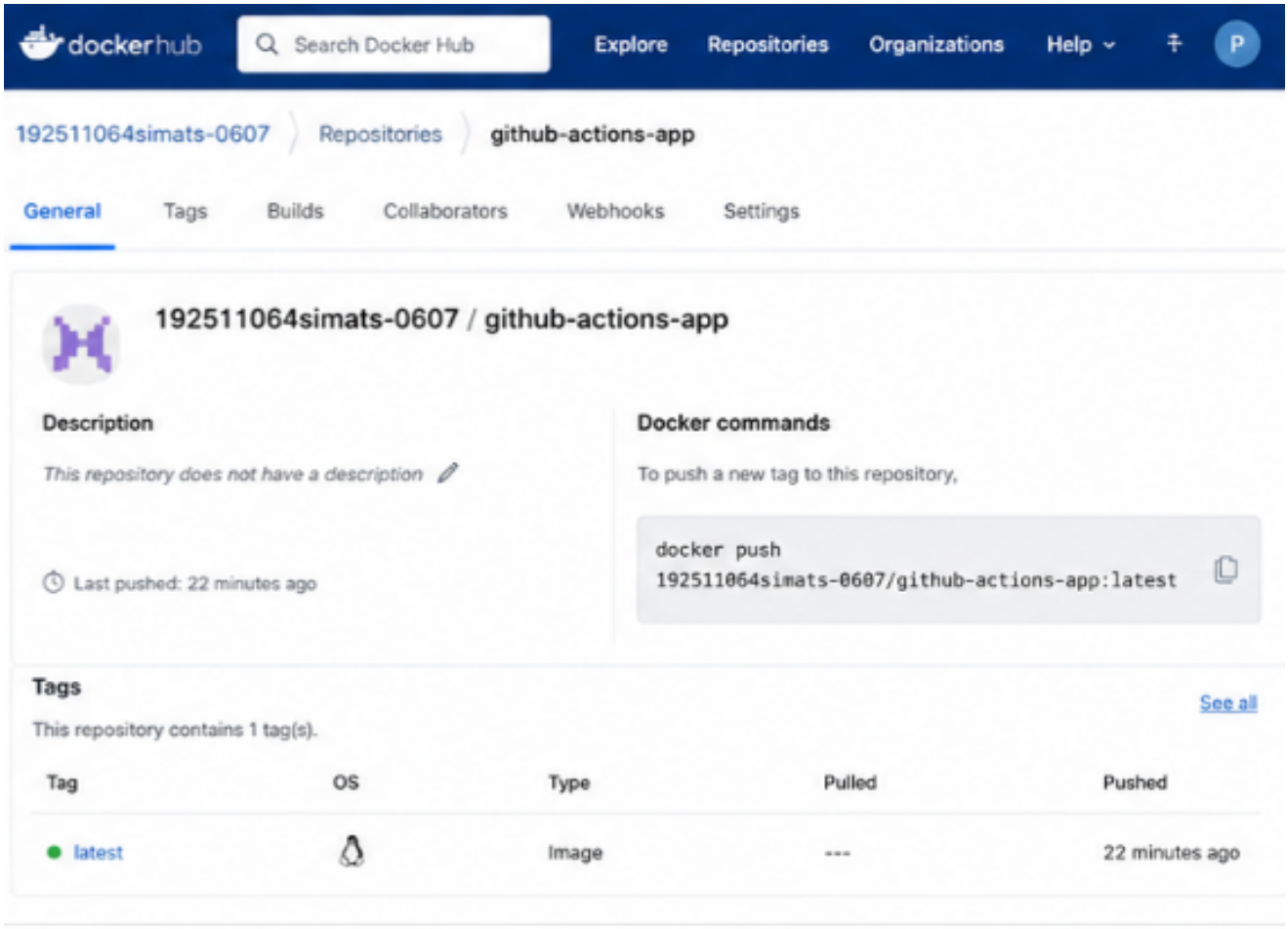
5. Test Continuous Deployment

Push a new code change to the main branch and monitor the Actions tab. Verify that the workflow completes successfully, the Docker image is updated, and the application is automatically redeployed.

Screenshot 1 – GitHub Actions Workflow Success


192511064simats-0607 / github-actions-docker
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Actions
New workflow
All workflows
Docker CI/CD Pipeline
Management
Caches
Runners
Docker CI/CD Pipeline
docker-deploy.yml
Re-run all jobs
This run has succeeded
main · c3d3e9f
Triggered via push
22 minutes ago
Status
Success
Total duration
2m 37s
Artifacts
–
docker-deploy.yml
on: push
Checkout code 2s
Build Docker Image 27s
Push Docker Image 18s
Deploy to Server 1m 25s

Screenshot 2 – Docker Hub Image



Verification

The GitHub Actions workflow shows a successful run with the Docker image build and push stages completed. The Docker Hub repository shows the latest image tag, confirming that the container image was successfully published.

Result

Thus, Continuous Deployment was successfully implemented using GitHub Actions and Docker. The application image was automatically built and pushed after code changes, demonstrating an automated CI/CD deployment workflow.